

Internet of Things

Learning Objectives

- Define IoT and identify its core architectural components
- Explain how sensors, connectivity, and data processing work together in IoT systems
- Design a basic IoT system for a practical application

Engage (5-7 minutes)

Ask students to count every internet-connected device in their homes. Beyond phones and laptops: smart speakers, gaming consoles, smart TVs, fitness trackers, security cameras, even refrigerators. Each device collects and shares data—that is the Internet of Things. Pose the question: what happens when everyday objects become “smart”?

Explore (10-12 minutes)

Students design a “smart classroom” IoT system. They identify: what data to collect (temperature, noise level, attendance, lighting), what sensors to use, how data travels (Wi-Fi, Bluetooth, cellular), where it’s processed (cloud vs. edge), and what actions result (adjusting AC, sending alerts). They sketch the system architecture.

Explain (10-12 minutes)

The Internet of Things connects physical devices to the internet, enabling them to collect, exchange, and act on data. The IoT architecture has four layers: perception (sensors detecting physical phenomena), network (connectivity via Wi-Fi, Bluetooth, LoRa, 5G), processing (cloud or edge computing analyzing data), and application (user interfaces and automated responses). IoT enables smart homes, wearable health monitors, industrial sensors, and smart city infrastructure. By 2030, an estimated 50 billion devices will be IoT-connected.

Elaborate (8-10 minutes)

Examine IoT communication protocols: MQTT for lightweight messaging, CoAP for constrained devices, HTTP for web integration. Discuss edge computing—processing data near the source to reduce latency and bandwidth. Analyze real IoT ecosystems: Amazon Alexa/Smart Home, industrial IoT (IIoT) for predictive maintenance, and smart agriculture using soil sensors. Address challenges: interoperability, power consumption, and firmware updates.

Evaluate (5-8 minutes)

Students design an IoT system for a specific scenario (elderly care, traffic management, or energy conservation). They must identify sensors, connectivity choices, data flow, processing decisions, and user interactions. Assessment: technical feasibility, appropriateness of component choices, and consideration of edge cases. Group critique of designs.